ENME 561/661: Mechatronics Design Lab

Fall 2026

Lab 1: MATLAB and Simulink

Submission: Individual

Due Date: Sep 17 , 11:59 PM

Submission Format: see the itemized checklist in "What to Submit" below (Section 3). Fill out the exit survey.

Important: Unlike other labs that are conducted in class, you are not required to be in the lab for this assignment. The lab space will remain closed, so complete this at home or any location with internet access.

> 🎓 **GRADUATE (ENME 661) REQUIREMENT — Look for the amber boxes**
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> If you are registered in ENME 661 (graduate section), you must also complete every item in an amber box like this one throughout the manual, in addition to all the standard tasks. These items are worth part of the same point total as the standard tasks (see the rubric in Section 4) — they are not extra-credit and skipping them will cost you marks even if the base task is done well. Undergraduate (ENME 561) students may read these boxes out of interest but are not required to complete them.

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1. Objective

MATLAB & Simulink are among the standard tools in mechatronics, robotics, automotive, aerospace, and industrial automation. In this course, we use MATLAB and Simulink as modelling and simulation tools to

analyze mechatronic systems, to solve engineering problems, to demonstrate problem-solving, to design, and to communicate technical findings using proper engineering tools and visualization. The goal of this lab is for students to familiarize themselves with these tools.

2. Preparatory Training (optional and highly recommended)

In case you are not familiar with MATLAB or Simulink, you are highly encouraged to complete:

- MATLAB Onramp[1] (estimated time: ~ 2 hours)
- Simulink Onramp[2] (estimated time: ~ 2 hours)
- Simulink Fundamentals[3] (estimated time: ~8 hours)
- MATLAB Live Script4 (20 minutes)

These tutorials are free, hands-on, and available through MathWorks. Sign in with your UCalgary email address — this is what gives you free access; if your UCalgary email is not recognized by the MathWorks site, contact IT (not the instructor, but please inform the instructor as soon as possible) to confirm your MathWorks academic account is active before the deadline, since a handful of students each term run out of time waiting on this step.

Completion is not graded and certificates are not submitted to D2L — they are for your own skill-building and resume.

> 🎓 **GRADUATE (ENME 661) REQUIREMENT — Additional preparatory step**
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> Skim the MathWorks documentation pages for fixed-step vs. variable-step solvers[5] before starting Task 1 (linked in Section 5, Task 1.7). No submission is required for this step — it is preparation for the graduate tasks below.

3. What to Submit (read this first)

Submit the following four items to the D2L Dropbox. Follow the file names exactly — this is the #1 source of grading delays.

File	Format & exact contents	Applies to
Simulink model	One .slx model file (the Simulink model itself — do NOT generate or submit C/C++ code with Simulink Coder). Name it LastName_FirstName_Q1.slx.	All students
Q1 write-up	One PDF (1–2 pages) with: Scope screenshots at the different frequencies you tested (e.g., original vs. at least one changed frequency for each signal) so the	All students

[1] https://matlabacademy.mathworks.com/details/matlab-onramp/gettingstarted
[2] https://matlabacademy.mathworks.com/details/simulink-onramp/simulink
[3] https://matlabacademy.mathworks.com/details/simulink-fundamentals/slbe
[4] https://www.mathworks.com/help/matlab/matlab_prog/what-is-a-live-script-or-function.html
https://www.mathworks.com/help/matlab/matlab_prog/create-live-scripts.html
[5] www.mathworks.com/help/simulink/ug/choose-a-solver.html

	reader can see what you're describing, plus a short-written observation of how varying each signal's frequency changed the result — the observation must reference and be supported by the screenshots, not stand alone. Name it LastName_FirstName_Q1_report.pdf.	
Q2 MATLAB Live Script	One PDF export of your .mlx file (File → Export → PDF in MATLAB), including code, plots, and written answers to the four reflection questions. Name it LastName_FirstName_Q2.pdf.	All students
Exit survey	Submitted through the Microsoft Forms link below, with your full legal name.	All students

4. Grading Rubric

Total: 100 points, identical weighting for ENME 561 and ENME 661. Graduate students earn the same 45/45/10 point allocations, but the graduate column below describes what is additionally required to earn full marks in each component.

Component	Points	Undergraduate (ENME 561) criteria	Graduate (ENME 661) — additional criteria for full marks
Q1 – Simulink model	45	Correct sine + square wave generation; correct sum logic (output = 1 when 4 ≤ sum ≤ 5, else sum); Scope screenshots included in the Q1 report that directly back up the reported observation (e.g., before/after captures at the different frequencies tested, not a single generic screenshot); frequency varied with a brief written observation tied to that evidence.	All undergraduate criteria, plus: A solver-choice justification (Task 1.7) with supporting evidence (plots, numeric comparison).
Q2 – ODE solver	45	Correct Forward Euler implementation; correct use of ode45 for verification; clear comparison plot; answers to the four reflection questions in the manual.	All undergraduate criteria, plus: empirical convergence study (Task 2.4) with a log–log error plot, an extracted observed order of convergence compared to theory, and a written stability/stiffness discussion (Task 2.5).
Exit survey	10	Submitted on time with full name included.	Same as undergraduate — no additional requirement.
Total	**100**		

5. Task 1 — Simulink Signal Generator and Scope (45 points)

Using Simulink, build a basic signal generator and scope:

1. Generate a sine wave with amplitude 2.
2. Generate a square wave with high and low values of +3 and –3.
3. Add the two signals together.
4. Configure the model so that if the resulting sum is between 4 and 5, the output is 1; otherwise, the output remains equal to the sum.

5. Display the output on a Scope.
6. Vary the frequency of each signal and observe/analyze the effect on the results. Include this observation in your Q1 report (Section 3).

> 🎓 **GRADUATE (ENME 661) REQUIREMENT — Task 1, additional graduate tasks**
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> **7. Solver justification.**
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> Run the same model with (i) a fixed-step solver and (ii) a variable-step solver, at two different step sizes each. Compare the four Scope outputs quantitatively (e.g., max error against the finest run) and write a short justification (150–250 words) for which solver/step-size combination you would recommend for this model and why.

6. Task 2 — Solving ODEs (45 points)

Write a MATLAB function that solves a linear ordinary differential equation numerically using Euler's method (Forward Euler). If you need to review the method, see Review-numerical simulation.pdf on D2L, or "Solving Differential Equations using Euler" (Chapter 17 of Hahn & Valentine, listed in Section 8) for a second worked treatment with MATLAB code.

Choose three different ordinary differential equations (they must not be identical). For each one, verify your function against MATLAB's built-in ode45[6] and compare the results. In your Live Script, answer:

1. State the three differential equations you chose. For each one, are the Euler and ode45 solutions the same? How do you compare them?
2. For each equation, which solution is more accurate, and why?
3. How can you increase the accuracy of the less accurate method?

> 🎓 **GRADUATE (ENME 661) REQUIREMENT — Task 2, additional graduate tasks**
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> **4. Convergence study.**
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> For one of the ODEs: Run your Forward Euler implementation at a minimum of five step sizes spanning at least two orders of magnitude (e.g., h = 0.1, 0.05, 0.01, 0.005, 0.001). For each h, compute the error against the ode45 solution (treated as the reference). Plot error vs. h on log–log axes, fit a line, and report the empirically observed order of convergence. State whether it matches the theoretical first-order (O(h)) behaviour of Forward Euler, and explain any discrepancy.
>
> **5. Stability / stiffness discussion.**
>
> For the ODE you were chosen, identify the largest step size h at which Forward Euler remains numerically stable (you may do this empirically by increasing h until the solution diverges/oscillates). Relate your observation to the ODE's characteristic root(s) and the known stability region of Forward Euler, in 150–250 words.

[6] https://www.mathworks.com/help/matlab/ref/function.html

7. Lab 1 Exit Survey — Reflection (10 points)

At the end of Lab 1, complete the online exit survey to reflect on your learning experience. It takes no more than one minute. Include your full name in the submission to receive credit. Submit before the lab deadline:

Lab 1 Exit Survey (Microsoft Forms)

8. Course Resources (Self-Study)

This course uses Quanser hardware (QUARC real-time control) in later labs, and Lab 1 asks you to implement a numerical ODE solver. Because students arrive with very different programming and numerical-methods backgrounds, the MATLAB/Simulink material below is not taught as in-class lecture content. Instead, use the resources below for self-study, and bring specific questions to office hours (Section 9). QUARC itself (Section 3 of the Primer) is not self-study - it requires the physical hardware and will be covered in the lab session where we start using it.

1. MATLAB/Simulink and QUARC Primer (Quanser Inc.). The file is posted to D2L. Covers the MATLAB environment and language, Simulink model-building, and the QUARC blocks used to run models on Quanser hardware in real time.

- You do not need to read it cover-to-cover before Lab 1 - Sections 1 (MATLAB) and 2 (Simulink) are the most relevant here.
- Section 4 is a one-page MATLAB command cheat sheet - worth bookmarking.

2. "Solving Differential Equations using Euler" — Chapter 17 (11 pages) of *Essential MATLAB for Engineers and Scientists*, by Hahn & Valentine. The file is posted to D2L. Directly supports Task 2 — walks through implementing Euler's method in MATLAB, which is the core of the reflection questions on accuracy and comparison to ode45.

- Office hours will focus on questions students actually bring — come with a specific block, error message, or concept (e.g., "why is my Euler solution diverging from ode45") rather than "explain Simulink," so we can use the time efficiently.

9. Getting Help

- Office hours
- MATLAB installation / MathWorks account issues: contact UCalgary IT, not the instructor, but inform the instructor about the issue ASAP— see Section 2.

10. Installing MATLAB on Your Personal Computer

Free installation through the university: https://www.mathworks.com/products/matlab/student.html

- Required: Simulink and the Control Systems Toolbox.
- Not recommended: installing all toolboxes (unnecessary disk space).
- Add-ons: any additional toolbox can be installed later via the Add-Ons menu in MATLAB.